

Part 2. Critical features of Part 1.

- The ways that quantum properties are useful to QUBIC researchers varies enormously.
- QUBIC needs a common language through which everyone can discuss how their work is quantum, AND how this differs to the classical world
- Knowing the difference drives experimental design and the language of publications
- The correspondence principle, through which quantum becomes classical as $\hbar \rightarrow 0$, is a unifying principle that provides such a language
 - The energy of a photon of light goes to zero, Maxwell's equations result
 - The energy of a quantum of vibration goes to zero, Newton's equations result
 - For IR and Raman spectroscopy, these quantum effects take the same form.
 - So classical mechanics provides a good approximation to quantum mechanics independent of $\hbar$.
 - Quantum effects simply diminish as $\hbar \rightarrow 0$
 - Effect of zero-point energy
 - Tunnelling reactions
 - Exception: vibrational entanglement, vibrational quantum qubits

Part 2. Critical features of Part 1.

- Rotational motion (gas-phase spectroscopy, nuclear spin e.g. diamond NV)
 - Extra complication as the number of conserved quantities changes:
 - Classical: x,y,z components of angular momentum are conserved
 - Quantum: only the total angular momentum and its z component are conserved
 - Also certain states become forbidden (overprescribed)
 - Eg, Bohr-Sommerfeld model of H atom
 - Classically: angular momentum is $0 \leq l \leq n$ for principle q.n. n
 - Quantum: $0 \leq l \leq n - 1$... no Bohr circle
 - Eg, rotational levels of a symmetric top
 - Classical: $2l + 2$ for total angular momentum l
 - Quantum: $2l + 1$... the two classical orbits adjacent to the separatrix interfere to merge into one state
 - The demanded uncertainty in quantum mechanics washes out the chaos of classical rotation !
 - As $\hbar \rightarrow 0$, more classical paths become accessible and chaos increases

Part 2. Critical features of Part 1.

- Electronic energy levels go to infinity as $\hbar \rightarrow 0$
 - Electronic transition energies (visible and UV spectra) go to infinity
 - But photon energies go to zero ... Opposite behaviour
 - massive changes to spectroscopic properties
- Chemical bonding changes as $\hbar \rightarrow 0$:
 - covalent bonding is lost with just a small reduction in $\hbar$
 - the dispersion interaction goes to zero smoothly
 - ionic bonding is not affected
- the Born-Oppenheimer approximation becomes exact
- Maxwell's equations give the lifetime of matter as of order 1 ns

Getting Quantum information from classical MD

- The trick is to use a formalism in which the quantum equations take the same form as the classical equations: Hamilton's equation
- We are interested in low frequency modes and so will use the classical temperature operator instead of the quantum one, so don't get scared.
- I give here the quantum equations and show how they reduce to the classical ones
- QUBIC is interested in non-equilibrium energy distributions, this is an easy addition
- I will give equations in terms of 1 degree of freedom only, but these readily generalise to include all possible motions of the atoms in a protein

Quantum thermal average in terms of eigenstates

- The quantum temperature operator is

$$\frac{e^{-\beta H}}{\text{Tr}(e^{-\beta H})}$$

where $\beta = \frac{1}{k_B T}$ with k_B as Boltzmann's constant, T the temperature, and Tr is the trace operator (which just averages over all wavefunctions)

- The value of any observable property operator A can be expressed as

$$\langle A \rangle = \frac{\text{Tr}(Ae^{-\beta H})}{\text{Tr}(e^{-\beta H})}$$

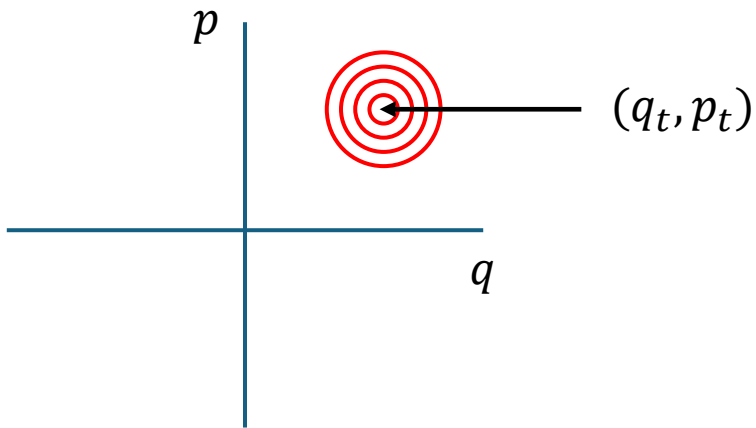
which can be expressed in terms of the eigenstates $|\psi_i\rangle$ of the system as

$$\langle A \rangle = \frac{\sum_i A_{ii} e^{-\beta E_i}}{\sum_i e^{-\beta E_i}}$$

Replace eigenstates with Gaussian wavepackets

- A gaussian wavepacket is a wavefunction that is:
 - is not an eigenstate, so it varies with time t
 - is expressed in terms of the position operator q and the momentum operator p ,
 - is centred in at position q_t and momentum p_t
 - has a phase γ_t
 - and a width α_t (spread in phase space so that it can obey the Heisenberg uncertainty principle)

$$|\psi_{q_t p_t}\rangle = \exp \left\{ \frac{i}{\hbar} [\alpha_t (q - q_t)^2 + p_t (q - q_t) + \gamma_t] \right\}$$



$$\langle q \rangle = q_t, \quad \langle p \rangle = p_t$$

Propagating a wavepacket forward in time

- The wavepacket advanced by time t under Hamiltonian operator H is

$$|\psi_{q_t p_t}\rangle = e^{-iHt/\hbar} |\psi_{q_0 p_0}\rangle$$

- If H is harmonic, then Gaussian wavepackets can change width

$$\frac{d\alpha_t}{dt} = -\frac{2\alpha_t^2}{m} - \frac{1}{2} \frac{\partial^2 V}{\partial q_t^2} \dots = 0 \text{ if } \alpha_t \text{ is set to match the force constant}$$

- and phase

$$\frac{d\gamma_t}{dt} = \frac{i\hbar\alpha_t}{m} - \frac{dS}{dt} \dots S \text{ is the classical action}$$

- with their centres evolving according to Hamilton's equation:

$$\frac{dp_t}{dt} = -\frac{\partial H}{\partial q_t} \quad \frac{dq_t}{dt} = \frac{\partial H}{\partial p_t}$$

- So there is a close relationship to classical mechanics
- For anharmonic potentials, assuming no change in wavepacket form:
 - Allowing the width to change with time is called the THAWED GAUSSIAN APPROXIMATION TGA
 - Constraining the width is called the FROZEN GAUSSIAN APPROXIMATION FGA

Quantum thermal average in terms of Gaussian wavepackets

$$\langle A \rangle = \frac{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 \langle \psi_{q_0 p_0} | A e^{-\beta H} | \psi_{q_0 p_0} \rangle}{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 \langle \psi_{q_0 p_0} | e^{-\beta H} | \psi_{q_0 p_0} \rangle}$$

- As $\hbar \rightarrow 0$, these Gaussian wavepackets collapse to delta functions at the classical points (q_0, p_0)

- This expression therefore collapses to

$$\langle A \rangle = \frac{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 A(q_0, p_0) e^{-\beta E(q_0, p_0)}}{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 e^{-\beta E(q_0, p_0)}}$$

where $E(q_0, p_0)$ is the total energy at that position and momentum.

- This is the **classical** equation used to determine properties using Monte Carlo simulations, which give the same results as does MD simulations

Preserving the form of the equations at finite $\hbar$

- Without approximation, the quantum temperature operator may be rewritten as a quantum time propagator:

$$\exp(-\beta H) = \exp\left(-\frac{iH\tau}{\hbar}\right) \text{ where } \tau \text{ is the imaginary time interval } \tau = -i\beta\hbar$$

- Hence the quantum temperature operator requires the solution of classical MD equations in imaginary time !
- We won't go there, just use the classical temperature approximation instead:

$$\langle A \rangle = \frac{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 e^{-\beta E(q_0, p_0)} \langle \psi_{q_0 p_0} | A | \psi_{q_0 p_0} \rangle}{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 e^{-\beta E(q_0, p_0)}}$$

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Calculating IR and Raman spectra

- For Raman, this applies to single-photon experiments only, not the paired photons used by Sergey Kruk, Irena Kabakova, etc.
- The quantum operator A involves $\frac{d\mu}{dq} q$ for IR, and $\frac{d\alpha}{dq} q$,
 - where μ is the dipole moment operator
 - and α is the polarizability operator.
- So to model Warwick's experiments, Callum will need to evaluate the polarizability derivatives, perhaps in Cartesian coordinates ?????
 - I think now we don't need to do this, see later ...

In both quantum and classical mechanics, 1-photon spectra can be expressed as a Fourier transform of an autocorrelation function

$$\sigma(\omega) = \int_{-\infty}^{\infty} C(t) e^{i\omega t} dt$$

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Evaluating the autocorrelation function

$$A = e^{-\beta H} e^{iHt/\hbar} \frac{d\mu}{dq} q e^{-iHt/\hbar} \frac{d\mu}{dq} q$$

- For constant $\frac{d\mu}{dq}$, the integrals $\langle \psi_{q_0 p_0} | A | \psi_{q_0 p_0} \rangle$ can be evaluated analytically as Gaussian integrals.
- Hence the autocorrelation function can be calculated by averaging over a classical thermal ensemble, and hence the spectrum.
- If the **wavepacket widths don't change with time** (Frozen Gaussian Approximation) and **classical temperature** is used, then we know the action S analytically. When the integrals are done, the quantum spectrum reduced to the classical one!
- This is why MD is so useful for understanding the properties and vibrational spectroscopies of molecules !
- This result is independent of the value of $\hbar$.

Warwick's vibrational entanglement experiment

- Warwick wants to cool ONE protein or membrane long-range vibration (frequency order 50 cm^{-1}) down its zero-point level $|\psi^0\rangle$
- Then excite it to one quantum of vibration $|\psi^1\rangle$
- These states can then be entangled to make an initial state

$$|\psi_0\rangle = a|\psi_0^0\rangle + b|\psi_0^1\rangle$$

- The question is as to whether or not this quantum entangled state can remain entangled as the protein/membrane vibrates.

Modelling quantum coherence

- If it moves coherently, then the propagating wavefunction $|\psi_t\rangle$ will retain a simple relationship with the initial wavefunction $|\psi_0\rangle$, as told by the autocorrelation function

$$\begin{aligned} C(t) &= \langle \psi_0 | \psi_t \rangle = \langle \psi_0 | e^{-iHt/\hbar} | \psi_0 \rangle \\ &= a^2 \langle \psi_0^0 | \psi_t^0 \rangle + b^2 \langle \psi_0^1 | \psi_t^1 \rangle + ab \langle \psi_0^0 | \psi_t^1 \rangle + ab \langle \psi_0^1 | \psi_t^0 \rangle \end{aligned}$$

- There are 3 ways in which this function can decay with time:
 - T1: (classical) energy is transferred from the high-temperature surroundings to heat up the cooled mode ... **property of E_t for the cooled mode**
 - T2: (classical) the phase of the vibration averaged over different initial positions and momenta for the surroundings becomes randomised ... **property of (q_t, p_t)**
 - T2: (quantum) the wavepackets loose phase **property of γ_t (the classical action)**
 - This term causes the lifetime of entangled states to go to zero as $\hbar \rightarrow 0$.
- Ignore the tricky bit (eg for molecular rotation): the quantum equations may demand an ensemble of classical trajectories, with the ensemble showing increased T2 (classical) times owing to the averaging out of chaos ...